

ARTIFICIAL INTELLIGENCE (AI) - PROJECT MILESTONE 2

Prompt Engineering for Video Human-Action Recognition using CLIP (ActionCLIP-inspired)

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1. Objective and Existing Work

The proposed project investigates prompt engineering for zero-shot human-action recognition. The research basis is **ActionCLIP: A New Paradigm for Video Action Recognition**, which formulates action recognition using video-text matching. In this implementation, a pretrained OpenAI CLIP model is applied to sampled video frames and candidate action prompts. The goal is to test whether different prompt formulations affect recognition scores. **Note:** this milestone implements the proposed prompt-engineering idea with CLIP on video frames; it is not a full reproduction of the original ActionCLIP training pipeline.

2. Environment and Implementation

Component	Configuration
Platform	Google Colab
GPU	NVIDIA Tesla T4, 15 GB
PyTorch	2.11.0+cu128
CUDA available	True
Model	openai/clip-vit-base-patch32 (pretrained)
Libraries	transformers, opencv-python

Test video: **0812.mp4** - 33.3 seconds, 60 FPS, 1,998 frames. Twelve evenly spaced frames were sampled to inspect the sequence. Approximate action segments were identified as: Swimming (0-7 s), Riding Motorcycle (10-17 s), and Playing Guitar (18-26 s).

3. Main Implementation Code

```
# Load pretrained CLIP
from transformers import CLIPProcessor, CLIPModel
model = CLIPModel.from_pretrained("openai/clip-vit-base-patch32").to("cuda")
processor = CLIPProcessor.from_pretrained("openai/clip-vit-base-patch32")
```

```
# Example action scoring
prompt = "a video of a person playing guitar"
inputs = processor(text=[prompt], images=frame,
                  return_tensors="pt", padding=True)
inputs = {k:v.to("cuda") for k,v in inputs.items()}
with torch.no_grad():
    output = model(**inputs)
score = output.logits_per_image[0,0].item()
```

The video experiment repeated the same scoring procedure over five sampled frames from each segment and averaged the resulting CLIP logits for each candidate action.

4. Experimental Results

Actual Segment	Swimming	Motorcycle	Guitar	Prediction
Swimming	31.9962	21.2584	19.1685	Swimming
Riding Motorcycle	21.3819	31.8518	21.4841	Riding Motorcycle
Playing Guitar	20.2970	20.8966	28.5698	Playing Guitar

All three segments were correctly classified in this controlled test: **3/3 = 100% segment-level recognition accuracy**. The highest similarity scores were 31.9962 for swimming, 31.8518 for riding a motorcycle, and 28.5698 for playing guitar. These are **CLIP similarity logits**, not probabilities.

5. Prompt-Engineering Experiment

For the original 39-second guitar video, four prompt formulations describing the same action were compared over eight sampled frames. The purpose was to test whether wording changes the image-text matching score.

Prompt	Average logit
a person playing guitar	27.3678
a person is playing guitar	27.8668
a video of a person playing guitar	28.7593
a person performing the action of playing guitar	27.0212

Best prompt in this test: “a video of a person playing guitar” with a mean logit of **28.7593**. This supports the hypothesis that prompt wording can affect CLIP-based action recognition scores. A larger multi-video benchmark is still required before making a general performance claim.

6. Discussion and Limitations

The experiment demonstrates a working video-to-text action recognition pipeline using sampled frames and pretrained CLIP. The controlled three-segment video produced 100% recognition for the selected candidate actions. However, the current evaluation is small and uses generated/controlled video rather than a full benchmark such as UCF101. The results therefore demonstrate feasibility, not state-of-the-art accuracy. The experiment also evaluates CLIP frame matching rather than reproducing the complete temporal encoder and training procedure of ActionCLIP.

7. Conclusion

The proposed prompt-engineering implementation was successfully executed in Google Colab using a Tesla T4 GPU and a pretrained CLIP model. Different prompt formulations produced different similarity scores, and the multi-action video was correctly classified in all three evaluated segments. The implementation provides a practical baseline for Milestone 3, where the best-performing prompt strategy can be expanded and evaluated on more videos/actions.

8. Research Sources

Wang, M., Xing, J., & Liu, Y. (2021). *ActionCLIP: A New Paradigm for Video Action Recognition*. arXiv:2109.08472. <https://arxiv.org/abs/2109.08472>

Radford, A. et al. (2021). *Learning Transferable Visual Models From Natural Language Supervision*. ICML 2021 (CLIP). <https://arxiv.org/abs/2103.00020>

ActionCLIP official implementation. <https://github.com/sallymmx/ActionCLIP>

Milestone 2 deliverables covered: completed-work description, implementation/run results, and result evidence. The Colab notebook contains the executable cells used for the experiment.